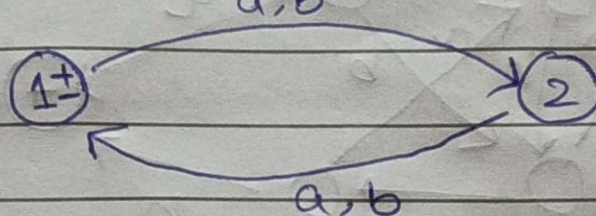


Example # 1

Even Number of letters

$$\Sigma = \{a, b\}$$

Only Need two states

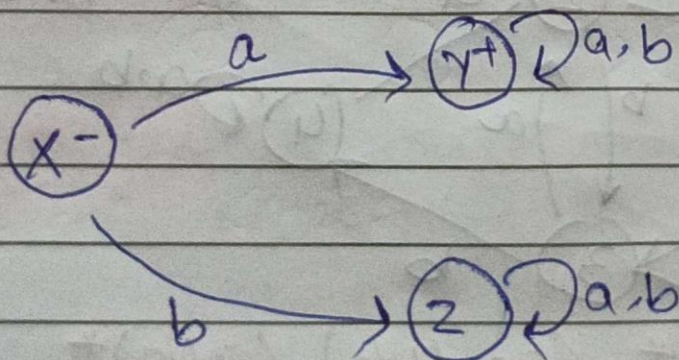


Strings: λ , aa, bb, ab, ba, aaaa, aabb, ...

Example # 2

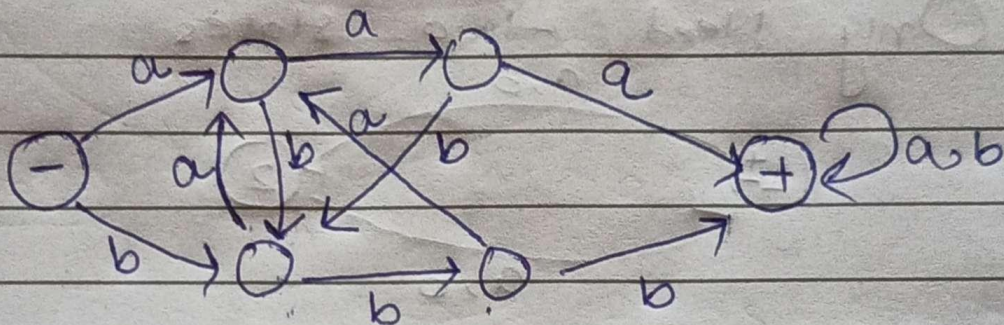
Strings that begin with letter a

$$a(a+b)^*, \quad \Sigma = \{a, b\}$$



Example # 3

Words containing triple letter
either aaa OR bbb (subset)



$$(a+b)^* (aaa + bbb) (a+b)^*$$

Example 4

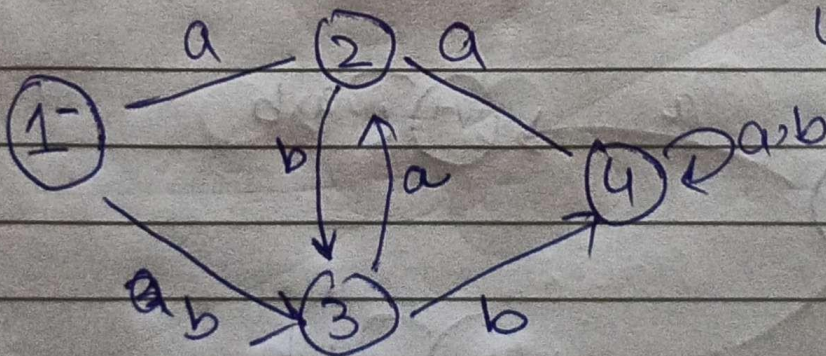
Words containing aa or bb
as a subset

e.g. ababaaab

1 → 2 → 3 → 2

4 ← 2 ← 2 ← 3 ← 1

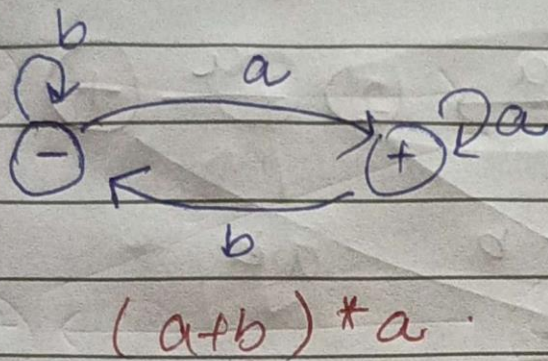
ababaa



$$(a+b)^* (aa + bb) (a+b)^*$$

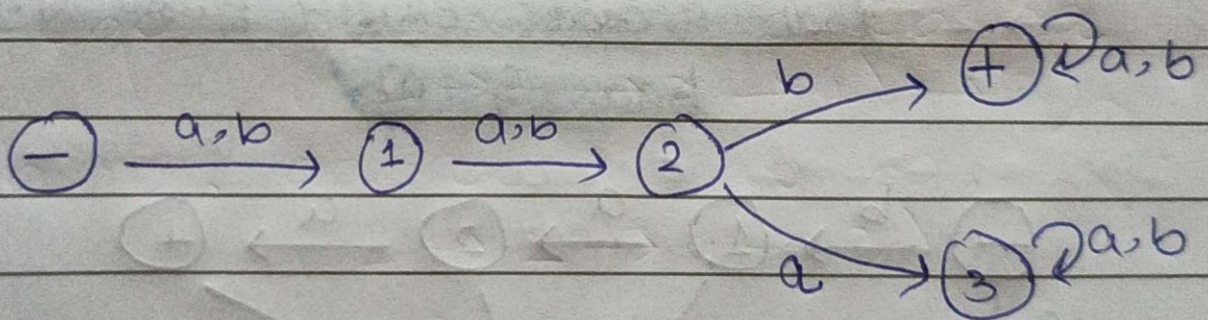
Example 6 :

FA accepts words that end on a



Example 7 :

FA will accept all words with b as third letter & reject all other words



$(aab + bbb + abb + bab)(a+ab)^*$

aab

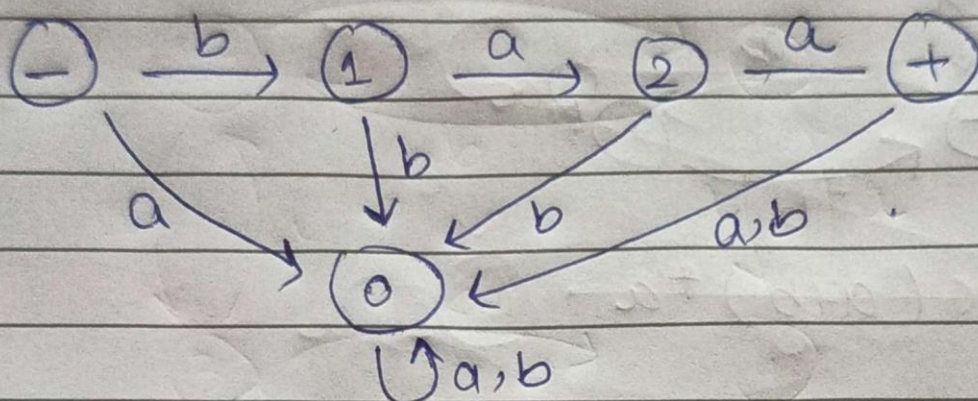
bab

bbb

abb **LUCKY**

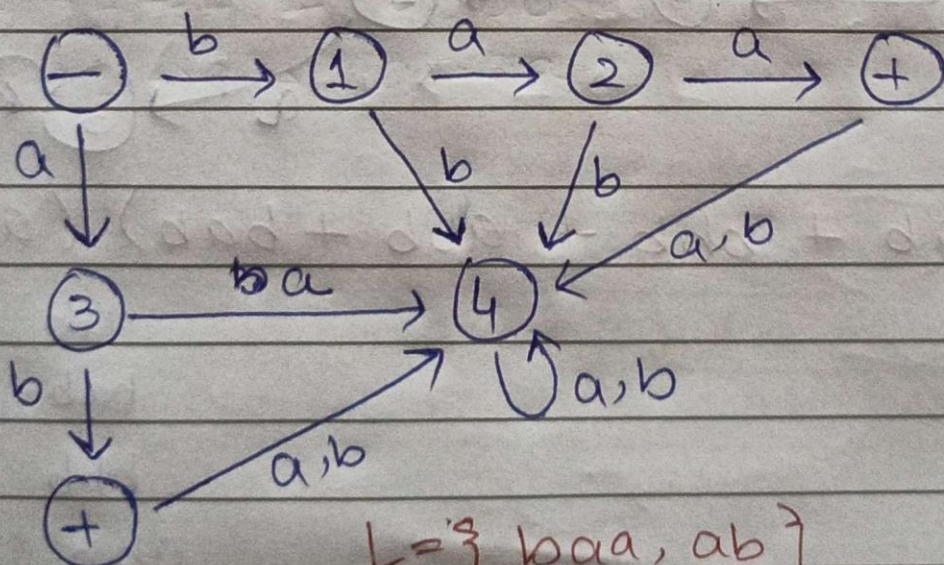
DATE

Example 8:

that accept only the word baa 

$$L = \{baa\}, R.E = baa$$

Example 9:

that accepts exactly two strings
 baa & ab 

$$L = \{baa, ab\}$$

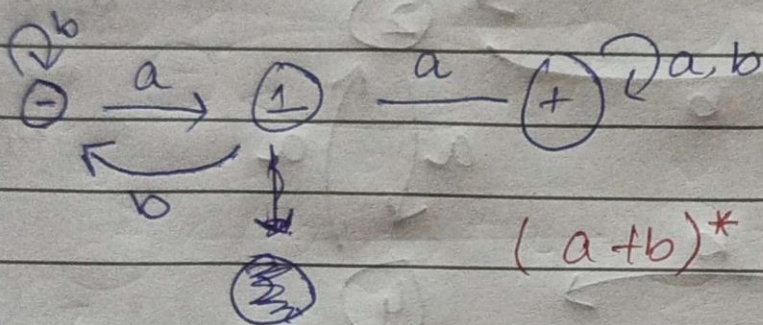
$$R.E = baa + ab$$

LUCKY®

Example 10:

All words with a double a in them

some where.



$$(a+tb)^* aa (a+tb)^*$$

